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Effects of Nanoscale Roughness on the Lubricious Behavior of an Ionic Liquid

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While many mechanistic studies have focused on the lubricious properties of ionic liquids (ILs) on ideally smooth surfaces, little is known about the mechanisms by which ILs lubricate contacts with nanoscale roughness. Here, substrates with controlled density of nanoparticles are prepared to examine the influence of nanoscale roughness on the lubrication by 1-hexyl-3-methyl imidazolium bis(trifluoromethylsulfonyl)imide. Atomic force microscopy is employed to investigate adhesion, hydrodynamic slip, and friction at the lubricated contact as a function of surface topography for the first time. This study reveals that nanoscale roughness has a significant influence on the slip along the surface and leads to a maximum slip length on the substrates with intermediate nanoparticle density. This coincides with the minimum friction coefficient at sufficiently small contact stresses, likely due to the lower resistance of the IL film to shear. However, at the higher pressures applied with a sharp tip, friction increases with nanoparticle density, indicating that the IL is not able to alleviate the increased dissipation due to roughness. The results of this work point toward a complex influence of the surface topology on friction. This study can help design ILs and nanopatterned substrates for tribological applications and nano- and microfluidics.

1. Introduction

Ionic liquids (ILs) are low-temperature molten organic salts that consist solely of anions and cations.[1] As a result of their asymmetrical shape, organic nature, and large molecular dimensions of, at least, one of the polyatomic ions, the intermolecular electrostatic interactions are weakened, and the conformational entropy is increased. All this hinders crystallization, and hence, ILs remain liquid over a wide range of temperatures in the absence of a solvent. Specifically, room temperature ILs, like the one used in this work, exhibit melting points below 100 °C. Since the field of IL boundary lubrication was initiated by Ye et al. in 2001,[2] ILs have gained growing interest in lubrication engineering because of their thermal stability, vanishingly low vapor pressure, electrical and thermal conductivity, and nonflammability.[3] Furthermore, their extraordinary chemical versatility has earned them the reputation as "designer" lubricants.[4] A key factor dictating the lubricious behavior of ILs is their ability to remain firmly adsorbed on surfaces at high contact stresses,[5,6] and thereby, to provide a plane of low shear strength, which mitigates friction and wear.[2]

Since the first surface forces apparatus (SFA) measurements by Horn et al. demonstrated that ethyl ammonium nitrate arranges in molecular layers between atomically smooth (negatively charged) mica surfaces,[7] many experiments and simulations have revealed a similar layered structure for other ILs at the solid–liquid interface. There is a number of works addressing different aspects of the behavior of ILs at interfaces,[5,8] including a recent chapter on the tribological behavior of ILs by two of the authors.[6] Besides the composition of the liquid,[9,10] other parameters such as surface chemistry and charge,[8,11] surface potential,[10,12,13] and humidity,[14,15] have been found to affect the interfacial structure of ILs.

The relation between the layered structure of the confined films and friction (and adhesion) appears as a "quantized" friction coefficient (and pull-off force), which increases stepwise with a decrease in the number of confined layers.[16] The increase in the number of layers is sometimes associated with more lubricious boundary films, and the increase in the force required to squeeze out the layers implies a more strongly adsorbed boundary film and higher cohesive forces and order. The investigation of the dependence of friction on velocity in smooth contacts revealed an intricate lubrication behavior.[17] Depending on the load, friction can increase, decrease, or remain constant with sliding velocity. This was explained through the strength and the dynamics of the interionic interactions at the slip plane via a stress-assisted thermally activated slip model. Molecular dynamics (MD) simulations[13] have also shown that the shift of the slippage plane from the solid/liquid interface to the interior of the film (upon an increase in surface potential in that cited work) leads to a notable decrease in

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the friction coefficient. More recently, friction measurements on stainless steel have revealed that friction and the degree of interfacial structure are inversely correlated.[18] This means that more ordered (structured) films leads to higher energy dissipation because stronger interlayer bonds need to be broken at the slip plane compared to disordered films. This suggests that ILs ordering with more difficulty, e.g., due to the incommensurate molecular structures between ions and a rough surface, could facilitate the rupture of interlayer bonds, and thereby, favor the decrease in friction. This hypothesis has motivated the present work. Interestingly, MD simulations have used this mechanism to describe the effect of topology on friction at rough contacts lubricated by ILs. Specifically, Mendonça et al.[19] showed that roughness (induced by conical protrusions) led to a decrease in the charge ordering of the IL film and thereby in friction; in contrast, David et al.[20] showed that the presence of periodic ridges and valleys caused an increase in IL ordering and friction.

We have recently reported an extensive study of the structural forces between two surfaces with roughness modulated via sintered nanoparticles in 1-hexyl-3-methylimidazolium bis(trifluoromethyl-sulfonyl)imide.[21] Although a smaller number of interfacial layers was detected on the rougher contacts, reflecting a higher disorder of the boundary films, interfacial layering was still resolved in force measurements by atomic force microscopy (AFM). However, it was shown that the interfacial composition and arrangement of the IL ions on rough substrates deviated from those observed on flat surfaces. Furthermore, we observed a reduction of the force required to squeeze out the layers located furthest from the hard wall (labeled as layer 3 in ref. [21]) with increase in roughness. In contrast, the force required to squeeze out the IL layer closest to the surface (layer 1) notably increased with nanoparticle density, using both, sharp tips and colloidal probes. This was hypothesized to result from the higher resistance of the liquid to flow through the tortuous contact, i.e., to stem from a hydrodynamic drag. In the present work, the lubrication of smooth and rough contacts by 1-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide has been investigated. Specifically, we have determined the effect of surface roughness on hydrodynamic drag, adhesion, and friction to provide insight into the mechanisms underlying lubrication.

2. Results

1-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide (abbreviated as [HMIM][TFSI]) was selected for this study. Figure S1 in the Supporting Information shows the size of anion and cation, as determined by molecular mechanics with the Avogadro software (Version 1.0.3). [HMIM][TFSI] was equilibrated at 44% RH (relative humidity) at room temperature for 10 days. The water uptake of [HMIM][TFSI] was determined gravimetrically to be ≈0.18 wt%, which agrees well with reported data obtained by coulometric Karl Fischer for this IL at ≈40% RH (110± 36 ppm, i.e., 0.19 wt%[22]). The viscosity of [HMIM][TFSI] has been reported to be 70 mPa s at 25.0 °C via precise viscosity measurements.[23] The rough substrates were prepared following the protocol in refs. [24–26] (see details in the Experimental Section). In brief, nanoparticles of 12 nm in diameter were adsorbed for 15 and 25 min on silicon wafers coated with a polycation to enhance electrostatic attraction and sintered afterward at 1080 °C using a heating ramp of 10 °C min^{-1}, followed by a cooling ramp of 2 °C min^{-1}. Bare silicon wafers (without nanoparticles) were sintered according to the same protocol and represent the substrate with a nanoparticle density of 0 μm^{-2}.

2.1. Surface Topography

The surface topography of the three silica substrates selected for these studies is shown in **Figure 1**, and the topographic details are summarized in **Table 1**. Note that the scale of the X and Y axes of the selected section profiles is different to ease the visualization of surface features. After sintering the substrates, the nanoparticle density was determined to be a) 0, b) 277 ± 5, and c) 721 ± 4 μm^{-2}, respectively. The substrates in (b) and (c) are labeled as 275 and 720 μm^{-2} substrates. The average distance between the nanoparticles is 85 ± 29 nm in (b) and 57 ± 21 nm in (c), and their respective root mean square (RMS) roughness is 4.9 ± 0.7 and 6.0 ± 0.5 nm. The RMS roughness of the silica substrates in the absence of sintered nanoparticles (0 μm^{-2} substrates) is 0.2 nm but values up to 0.5 nm were measured after sintering.

2.2. Pull-Off Force Measurements and Multiasperity Contact Geometry

Figure 2 shows representative load-dependent pull-off force measurements on substrates with a nanoparticle density of 0 (circles), 275 (diamonds), and 720 μm^{-2} (triangles) immersed in [HMIM][TFSI] using a sharp silicon tip (radius $R = 48$ nm). The measurements were conducted at an approach velocity of 30 nm s^{-1} to eliminate hydrodynamic effects. For instance, the pull-off force measured with an AFM tip at the applied normal load of 7.5 nN is 0.5 and 0.6 nN for particle densities (ρ) of 0 and 275 μm^{-2}, respectively, whereas it is higher, ≈1.7 nN, for $\rho = 720\ \mu m^{-2}$.

The increase of the pull-off force with a change of surface topography is usually associated with an increase in the true contact area.[27] The contact radius (a) was estimated based on the Derjaguin–Muller–Toporov (DMT) contact mechanics model for ideally smooth materials that undergo elastic deformation. To determine the adhesion energy required for the DMT model (see the Experimental Section), it was assumed that only van der Waals interactions are responsible for the measured pull-off force. A Hamaker constant of $A_H = 6.7 \times 10^{-21}$ J (calculated with a refractive index of 1.43, 1.448, and 3.887 and a relative permittivity of 12.7, 3.8, and 15.112 for the IL, silica substrate, and silicon tip, respectively) is obtained for this IL sandwiched between the tip and substrate, which leads to a surface energy $\gamma = -A_H/24\pi(\frac{\sigma}{2.5})^2 \approx 2.26$ mJ m^{-2}, with σ the diameter of an ion pair assumed here to be 0.5 nm (Figure S1, Supporting Information). Using this value for the interfacial energy, both the contact radius a and the mean contact stress (σ_0) were estimated for 0 μm^{-2} substrates and on single nanoparticle.

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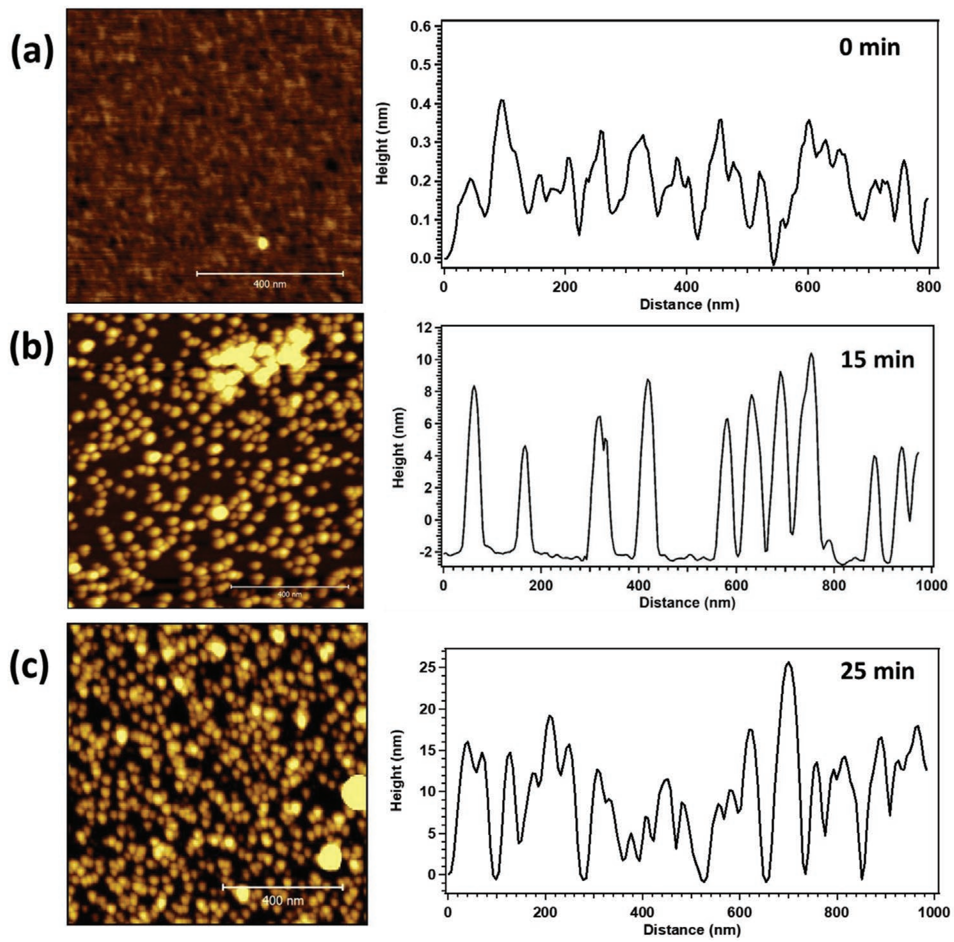

Figure 1. Representative AFM topographical images (1 µm × 1 µm) and corresponding cross-sections of the selected silica substrates with surface densities of a) 0, b) 277 ± 5, and b) 721 ± 4 µm^{-2}. AFM images were taken in air, in AC mode. The nanoparticles appear larger than they are because the tip cannot perfectly follow the topography of the nanoparticles. Note that the scale of the *Y*-axis of the section profiles is different for each immersion time to ease the visualization of the surface features.

Table S1 and Figure S2 in the Supporting Information show the calculated contact stress and contact area, respectively. This estimation helps us reconstruct the geometry of the multiasperity contacts. For example, a tip with a radius of 48 nm can come in contact with at most two nanoparticles on 275 µm^{-2} substrates and five nanoparticles on 720 µm^{-2} substrates; in the former, the tip can also probe the space between the nanoparticles (interparticle distance ≈85 nm, Table 1). The contact area of the tip with five nanoparticles is larger than with a perfectly smooth substrate.

The relation between pull-off force and load shown in Figure 2 reveals, however, a deviation from the DMT model, which predicts that the pull-off force is independent of load.[29] First, the presence of the confined liquid affects the distribution of the traction between liquid and solids, which depends on the ratio between the contact radius and the thickness of the confined film; increasing ratios imply a re-distribution from the liquid to the solid. Assuming a film thickness that would decrease from ≈2 to ≈1 nm with an increase of load from 2 to 10 nN, this ratio would vary from ≈0.3 to ≈2.0. Due to the quantization of adhesion at contacts lubricated by ILs,[16] the pull-off force could increase stepwise with load if the increase of load would lead to a decrease in the number of layers. In fact, abrupt jumps of the pull-off force are observed for the three substrates in the range of investigated loads (see arrows). In addition to the effect of the liquid, an increase of pull-off force with normal

Table 1. Topographical characteristics (nanoparticle density, average RMS roughness, peak-to-valley distance, and average lateral interparticle distance) obtained on at least ten profiles per image over three images.

Substrate	Immersion time [min]	NP density [µm^{-2}]	RMS roughness [nm]	Peak-to-valley distance [nm]	Asperity distance [nm]
0 µm^{-2}	0	0	0.13 ± 0.07	0.53 ± 0.04	NA
275 µm^{-2}	15	277.1 ± 4.6	4.85 ± 0.73	14.79 ± 3.35	84.72 ± 29.47
720 µm^{-2}	25	721.0 ± 3.9	6.03 ± 0.51	16.95 ± 4.35	57.25 ± 21.02

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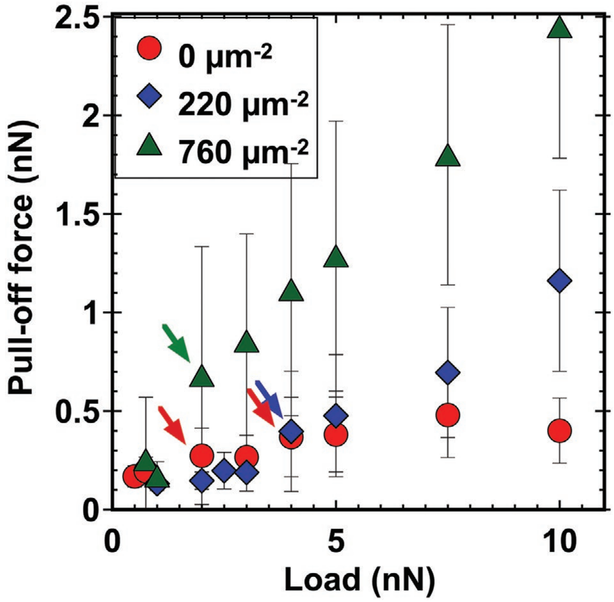

Figure 2. Load-dependent pull-off force measurements in [HMIM][TFSI] carried out with a sharp silicon tip. The large error bars suggest that the tip does not always come in contact with the same number of nanoparticles, and therefore, the contact area and the pull-off force vary. The maximum contact stress on the 0 µm^{-2} substrate is estimated to be 0.75 GPa (at 10 nN), while it is 3.47 GPa on a single nanoparticle. Yield strength of silica is 5.43 GPa.[28] Tip radius R = 48 nm.

load can result from elastoplastic and plastic deformation of nanosized asperities.[27,30,31] The relation between pull-off force and normal load depends on the mode of rupture (ductile or brittle).[30] Figure 2 shows that the pull off force scales linearly with load on 720 µm^{-2} substrates, which agrees with the expected relation for the adhesion in the ductile mode. According to Maugis and Pollock theory, if a load larger than ≈52 nN is applied on a single nanoparticle with this sharp tip, the stress on a single nanoparticle will overcome the yield strength of silica (≈5.43 GPa).[28] While the applied load in this experiment is much smaller, the roughness of the nanoparticles could promote elastoplastic deformation at smaller loads. This specific roughness effect is, however, out of the scope of this work and was not further investigated.

When the colloid approaches to the substrate, it comes in contact with at most five nanoparticles on both 275 and 720 µm^{-2} substrates. The pull-off force measured with a silica colloid with a radius of ≈5 µm at an applied load of 7.5 nN is 2.74 ±0.20 nN for the smooth substrate and 1.57 ± 0.15 and 1.2 ± 0.11 nN for 275 and 720 µm^{-2} substrates, respectively. The higher adhesion on the 0 µm^{-2} substrates would suggest (for a nonlubricated system) that the contact area between the colloidal probes used in this work (diameters between 5 and 10 µm) and 0 µm^{-2} substrates (as calculated by the DMT model) is greater than on 275 and 720 µm^{-2} substrates. However, the pull-off force is also influenced by the quantized structure of the IL film, as described earlier. The required load for elastoplastic deformation is estimated to be ≈480 nN in multiasperity contacts with five nanoparticles, which is not applied in this work.

2.3. Friction Force Measurements by Colloidal Probe AFM

2.3.1. Smooth Surfaces

Figure 3a shows the friction force as a function of the applied normal load on the three substrates measured with a silica colloid by AFM. On the 0 µm^{-2} substrates, the friction force did not increase linearly with load, thereby deviating from Amonton's law. The deviation appeared as two different linear relations between friction and load and was confirmed in several replica experiments. At loads smaller than ≈10 nN corresponding to a contact stress of σ_0 ≈ 15 MPa, the friction coefficient (μ) was extremely small (μ < 0.003 (–)), while it increased to ≈0.058 (–) above 15 MPa, consistent with reported results for other imidazolium ILs on atomically smooth mica as substrate.[15] This suggests that the friction coefficient is quantized due to the

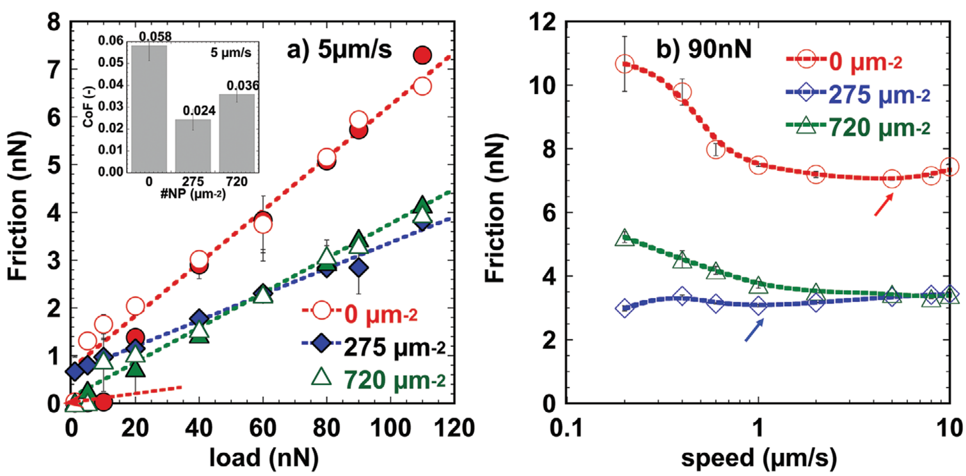

Figure 3. Low-pressure (colloid) friction measurements. a) Load-dependent friction force on silica substrates with nanoparticle densities of 0, 275, and 720 µm^{-2} at a sliding velocity of 1 µm s^{-1} measured with a silica colloid. The maximum contact pressure is σ_0 ≈ 63 MPa on the smooth substrate. The inset shows the corresponding friction coefficients. b) Speed-dependent friction force on the three substrates at a constant load of 90 nN (σ_0 ≈ 57 MPa). Based on the contact stress on the smooth substrate, these friction force measurements correspond to regime I in Figure 4a. Colloid radius = 5 µm.

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stepwise decrease in the number of layers confined between the two surfaces with an increase in load.[16] In some replica experiments, however, the low-friction regime was not observed and only a single slope was detected (see empty circles, Figure 3a), which reflects that the composition of the confined IL film remained constant upon an increase in load. Our extensive studies[15,17] of the influence of traces of water on the interfacial structure and on the friction force at mica-silica contacts have demonstrated that a higher water content at the interface can reduce the prominence of the low friction regime or even eliminate it, as observed here.

2.3.2. Rough Surfaces

The friction measurements with a silica colloid reproducibly confirmed the reduction of friction on the rough substrates (see inset in Figure 3a, for a sliding velocity of 1 µm s^{-1}). This decrease in friction extended over a wide range of sliding velocities (Figure 3b). This demonstrates that [HMIM][TFSI] is able to efficiently lubricate rough contacts with nanoparticle densities of 275 and 720 µm^{-2} despite the high stress applied on a single nanoparticle (σ_{NP} up to 1.25 GPa, Table S2, Supporting Information). Importantly, the friction coefficient was smallest on 275 µm^{-2} substrates over a wide range of sliding velocities, where friction was observed to remain approximately constant and to slightly increase with velocity at the highest sliding speeds (see arrow in Figure 3b), likely announcing the transition into hydrodynamic lubrication; a slight increase of friction with velocity was also observed on 0 µm^{-2} substrates at the highest velocities (see arrow). However, a decrease in friction with velocity was most prominent on 0 and 720 µm^{-2} substrates over a wide range of velocities. A decrease in friction with sliding speed has been observed also for other imidazolium ILs in colloidal probe AFM experiments (e.g., on mica[15,17]) and it has been associated with the slow relaxation of the confined ions: as the contact time during the sliding of the colloid becomes gradually smaller with increase in velocity, the confined ions have less time to interact with each other at the slip plane, which reduces the energy dissipated, and thereby friction.

The influence of nanoscale roughness on pull-off force and friction has been investigated on similar substrates but with a polyethylene colloid, yielding a strongly adhesive contact.[24,25] The cited works showed that friction is proportional to the pull-off force and to the contact area, and hence, if the nanoparticle density reduces the contact area, friction should decrease. Based on this, the reduction of the contact area with nanoparticle density inferred from the pull-off force measurements with the silica colloid should alleviate friction. However, this cannot explain the minimum of the friction coefficient on 275 µm^{-2} substrates, which suggests the contribution of an additional mechanism that is discussed later. We also propose that the discrepancy between our experimental results and those in ref. [24] originates from the weakly adhesive nature of the silica–silica contact lubricated by [HMIM][TFSI], compared to the strongly adhesive characteristic of the silica–polyethylene contact in water in ref. [24].

2.4. Friction Force Measurements with a Sharp AFM Tip

To examine the lubrication mediated by [HMIM][TFSI] at higher applied stresses, the frictional characteristics of the same substrates were also investigated using sharp tips. **Figure 4**a shows the friction versus normal load for the three selected substrates measured at a sliding velocity of 1 µm s^{-1}. The normal load ranged between 2 and 60 nN, which corresponds to an average contact stress σ_0 from 0.34 to ≈1.45 GPa on a smooth surface, well above the contact stress applied with the silica colloid. Replica experiments under the same experimental conditions showed very similar responses (see various measurements

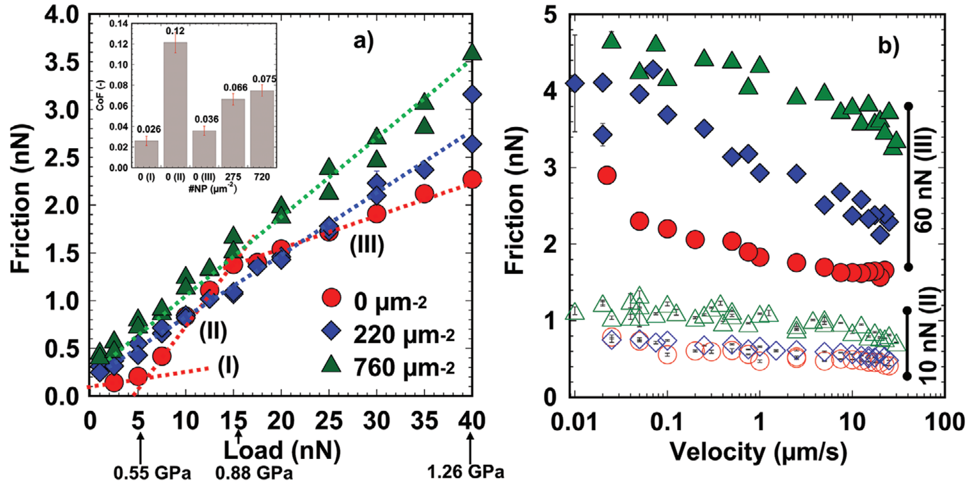

Figure 4. High-pressure (sharp tip) friction measurements. a) Friction versus normal load corresponding to an average contact stress σ_0 smaller than 1.26 GPa on a smooth surface and 5.55 GPa (at 40 nN) on a single nanoparticle at 1 µm s^{-1} and b) friction versus sliding velocity at two selected loads (10 and 60 nN, corresponding to σ_0 equal to 0.75 and 1.45 GPa on a flat surface and 3.47 and 6.36 GPa on a single nanoparticle, respectively). The inset in (a) shows the friction coefficient for a sliding velocity of 1 µm s^{-1}. Radius of tip R = 47.8 nm. The error bars are often smaller than the marker size and therefore not visible.

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in Figure 4a), indicating that the sample and the tip were not affected by wear during the experiment. The inset in Figure 4a summarizes the friction coefficients calculated as the slope of the friction versus load curves in the different linear regions, and as a function of the nanoparticle density.

2.4.1. Smooth Surfaces

As illustrated in Figure 4a, the friction force is affected by the nanoscale roughness, but the influence is intricate: three different linear regions (labeled as I, II, and III inspired by ref. [32]) were identified on 0 μm^{-2} surfaces as the load was increased, whereas a single linear relation was reproducibly measured on 275 and 720 μm^{-2} substrates. On the 0 μm^{-2} substrates, the friction coefficient in regime I was the smallest (≈0.026 (–)). The transition to regime II happened upon an applied normal load of 5 nN ($\sigma_0 \approx 0.55$ GPa), and hence, at a contact stresses not achieved with the silica colloid. The friction coefficient was increased by a factor of five to 0.12 (–) in regime II, and it decreased to 0.036 (–) at a contact stress above 0.88 GPa in regime III. Figure 4b demonstrates that friction was decreased with sliding velocity in regimes II and III. As described earlier, the decrease of friction with velocity can be justified by the slow relaxation of the confined ions, which dictates the intermolecular interactions at the slip plane. The velocity dependence in regime I was not investigated with the AFM tip.

The change of the coefficient of friction with load is intriguing. Three friction responses (labeled as regions I, II, and III) were also identified on a silica colloid-mica contact lubricated by ethyl ammonium nitrate.[33] In the cited work, the coefficient of friction, calculated as the slope of the friction versus load curves within each of the three regions, was shown to decrease from region I to region III. Based on normal force measurements, the region I (lowest loads) was associated with multilayer slip, whereas regime II and III were related to the lubrication provided by a cation monolayer strongly adsorbed on mica to neutralize its surface charge. The authors inferred from the velocity dependence of friction that the adsorbed monolayer was not fully compressed in region II, and therefore, relatively rough. A small structural transition happened upon a further increase in load (region III), and the dependence on the sliding velocity was seen to vanish. The authors concluded that the high pressure led to a smoothening of the adsorbed cation layer, and thereby to a decrease in the friction coefficient.

The small friction coefficient in regime I is consistent with the lubrication mediated by multilayer slip on smooth surfaces, like also invoked for the measurements with the silica colloid. The friction coefficients measured with silica colloids and sharp tips in regime I are slightly different. This is, however, not surprising since the roughness and chemical composition of the tip and colloid are different. The colloid is made of silica whereas the sharp tip is made of silicon with an oxide layer of a few nanometers that can be worn out easily in friction force measurements.[34]

It also seems reasonable that an increase in the pressure applied with the tip smoothens the surface-adsorbed IL ions, thereby justifying the decrease in the friction coefficient from regime II to III. Although the pressures applied with the sharp tip are much higher than in ref. [33] and hence, the adsorbed monolayer could be removed with the sharp tip, it seems unlikely that the removal of the IL film could induce a reduction of the friction coefficient compared to the lubrication mediated by the IL in regime II. Therefore, a smoothening, or an alternative structural change of the surface-adsorbed IL, is more likely to happen in regime III.

2.4.2. Rough Surfaces

As illustrated in Figure 4a, the presence of the nanoparticles on the substrate led to a single friction coefficient across the entire range of loads, in agreement with Amonton's law ($\mu \approx 0.066$ and 0.075 (–) for 275 and 720 μm^{-2} substrates, respectively). Taking into account the decrease in the number of detected IL layers induced by the nanoparticles,[21] it is reasonable that the lubrication by multilayer slip (regime I) is disturbed (and eliminated) by the presence of the nanoparticles. A comparison of the friction coefficient at loads higher than 15 nN shows that the friction coefficient increases with roughness, illustrating the limitation of this IL to lubricate the rough contact as efficiently as the smooth contact in the selected range of high stresses.

2.5. Hydrodynamic Force Measurements and Slip Length

The hydrodynamic drag upon approach of a silica colloid to the substrates was measured by colloidal probe AFM at approach velocities ranging between 0.5 and 20 $\mu m\ s^{-1}$; see the Experimental Section for more details. Representative measurements of the hydrodynamic force as a function of the surface separation (D) are shown in **Figure 5** for 0, 275, and 720 μm^{-2} substrates. The hydrodynamic drag increased with an increase in the approach velocity on the three substrates, as expected, but this increase was clearly less prominent on the 275 μm^{-2} substrates; note that the smallest hydrodynamic drag was measured at each specific velocity on this substrate.

To obtain more insight into the slip boundary condition of [HMIM][TFSI] along the selected surfaces, the experimental results were compared to existing models. Based on Reynolds lubrication theory,[35] the hydrodynamic force under no-slip boundary condition exerted on a sphere of radius R approaching a flat surface perpendicular with a velocity V is approximately given by

$$F_{\text{hyd}} = \frac{6\pi\eta R^2 V}{D} \tag{1}$$

where η is the bulk dynamic viscosity and D is the surface separation. The error in the model is insignificant when D is much smaller than R, a condition satisfied in this system. To determine the viscosity under the conditions of our experiments, we calculated the damping function $G = 6\pi R^2 V / F_{\text{hyd}}$ at large separations following ref. [36]. The slope G versus D gives the inverse of the viscosity of the liquid η. This yields 70 ± 2.7 mPa s, which agrees well with the reported value.[23] The average value of 70 mPa s was assumed in following calculations.

The black line in the three diagrams represents the calculated hydrodynamic drag assuming no-slip for selected approach

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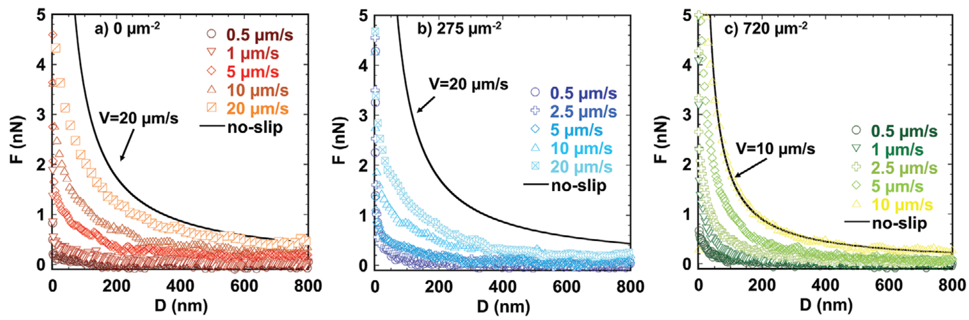

Figure 5. Hydrodynamic force measured with a silica colloid ($R = 4.0$ μm) approaching the 0, 275, and 720 μm^{-2} substrates in [HMIM][TFSI] at velocities ranging between 0.5 and 20 μm s^{-1}. The peak-to-valley distances in 0, 275, and 720 μm^{-2} substrates are 0.53 ± 0.04, 14.79 ± 3.35, and 16.95 ± 4.35 nm, respectively.

velocities (20 μm s^{-1} in (a) and (b) and 10 μm s^{-1} in (c)). As evidenced here, this calculation overestimates the hydrodynamic drag on 0 μm^{-2} substrates (also at other approach velocities), which is consistent with the slip of [HMIM][TFSI] at the solid boundary. On 275 μm^{-2} substrates, the no-slip boundary condition (black line) also overestimates the measured hydrodynamic drag, and the deviation is even more prominent than on 0 μm^{-2} substrates. In contrast, on 720 μm^{-2} substrates, the assumption of no-slip yields a better agreement between the model and the experiment at the selected velocity.

To account for the slip along the solid boundary, the model by Vinogradova[37] was used to calculate the hydrodynamic drag force

$$F_{\text{hyd}} = f^* \frac{6\pi\eta R^2 V}{D} \tag{2}$$

where f^* is the correction factor for slip for two surfaces with the same slip length (b), i.e.,

$$f^* = \frac{D}{3b}\left[\left(1+\frac{D}{6b}\right)\ln\left(1+\frac{6b}{D}\right)-1\right] \tag{3}$$

Although it is expected that the silica colloid and the substrates have different slip lengths, the same colloid was used for the three substrates, and hence, Equation (3) can be used to compare the results across the substrates.

The analysis of results acquired on structured surfaces like ours is complicated by the uncertainty in the boundary position ($D = 0$). Reliable slip lengths have been obtained by shifting the boundary downward, i.e., "within" the nanoparticles.[38] Placing the hard wall at the peak of the asperities can cause an overestimation of the slip length in the order of the peak-to-valley distance; in our case, less than 17 nm (Table 1). In addition to this, the presence of molecular layers at the solid/liquid interface, like in the case of ILs, introduces an additional interference in the analysis of the slip length. The liquid structure can lead to an additional shift of the boundary. For example, SFA measurements showed that approximately two layers of IL ions stick to mica and remain effectively immobilized during hydrodynamic force measurements ($V \approx 0.5$–1 nm s^{-1}, $R \approx 2$ cm).[39] Because of the overlap of these two phenomena and the uncertainty with regard to the liquid structure in our system, we have preferred not to shift the boundary for the analysis of the slip length. Nevertheless, we recognize that the estimated slip length can deviate from the true value by a few nanometers. This approach leads to good fits of the model to the experimental results at separations larger than ≈20 nm.

Figure 6 shows the slip length obtained from the best fit of Equations (2)–(3) to the experimental data. The increase in nanoparticle density from 0 to 275 μm^{-2} led to an increase of the slip length, which reflects the weaker hydrodynamic force in Figure 5. A further increase in the particle density to 720 μm^{-2} led to a prominent decrease in the slip length, which achieved values closer to the no-slip boundary condition.

The nonmonotonic change of the slip length with nanoscale roughness indicates that there are competing mechanisms underlying the slip condition. For interfaces characterized by weak solid–liquid interactions, the minimum surface roughness to achieve a no-slip boundary condition was shown to be ≈6 nm by Zhu and Granick,[40] which is very similar to the RMS roughness of the 720 μm^{-2} substrates (Table 1). That study showed that a surface roughness below 6 nm RMS led

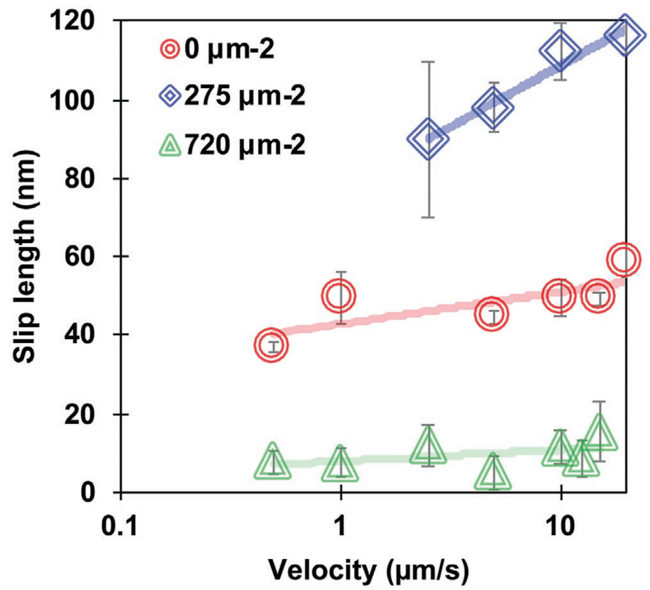

Figure 6. Slip lengths of [HMIM][TFSI] along 0, 275, and 720 μm^{-2} substrates as a function of the approach velocity. The same colloid was used for all measurements. Colloid radius = 4.0 μm.

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to considerable slip lengths, with the largest values reached on the atomically smooth surfaces. The results were thus in agreement with the argument that the viscous dissipation as the fluid flows past surface irregularities brings it to rest. The slip length on 275 μm^{-2} substrates (RMS = 4.85 nm) is, however, significantly larger than on the smooth substrates. Hence, in contrast to Zhu's work, the interactions between the surface and [HMIM][TFSI] and its interfacial structure must play a key role and it must be strongly influenced by the presence of the nanoparticles. Such influence of the topology on the interfacial structure was, in fact, demonstrated in our earlier work.[21] The size of the IL layer closest to the surface decreased from 8.1 to 5.1 Å with an increase in nanoparticle density while the force to squeeze-out this IL layer significantly increased. The roughness effect outcompetes the relevance of the interfacial phenomena and leads to a prominent decrease of the slip length on 720 μm^{-2} substrates. Note that the results were reproducible in replica experiments with a different colloid, and hence, we can exclude that the colloid is responsible for these results.

3. Discussion

Our experimental studies reveal two different scenarios depending on the applied pressure, which points toward a complex dependence of the friction coefficient on surface topology. In the range of small pressures, the friction coefficient was found to be smallest on the 275 μm^{-2} substrates (see, e.g., Figure 3a,b). MD simulations have shown that friction at contacts lubricated by ILs can both increase[20] or decrease[19] with nanoscale roughness, and have associated this change with the enhanced order or disorder of the ions in the confined films, respectively. Our measurements of the interfacial structure revealed a smaller number of well-organized interfacial layers with an increase in roughness, suggesting the enhanced disorder of the multilayer films.[21] Hence, one potential explanation is that the more disordered multilayer films on the rough substrates are responsible for the decrease in friction, as inferred from MD simulations.[19] However, this would lead to the lowest friction force on 720 μm^{-2} substrates, in contradiction to our results. The hydrodynamic force measurements thus serve to complete the picture about the lubrication of rough contacts by [HMIM][TFSI]. Note that the minimum hydrodynamic drag during the interfacial flow of the IL was attained on 275 μm^{-2} substrates (Figure 5). This suggests that the ease of this IL to slip along these substrates promotes lubrication and reduces the friction coefficient. This lubrication mechanism is, however, only efficient at sufficiently low pressures, since measurements with sharp tips clearly deviate from this behavior.

Despite the enhanced disorder of the confined IL films, a greater force was required to squeeze out the layer closest to the hard wall on the rough substrates compared to the smooth one.[21] Such an increase in the force could be justified by the higher order of this layer, which would explain the increase in friction with roughness, based on the cited MD simulations. Note that David et al. also found[20] that the molecules lying deeper in the valleys between ridges are more ordered and act as a template for subsequent ion layers. In addition to this effect, an increase in friction with roughness can be expected based on the asperity interlocking model.[27] Here, energy is needed to overcome the compressive external load and the attractive intermolecular interactions as the tip rises and slides across the asperities of the substrate. It is also worth mentioning that the stress at the contact between the sharp tip and a single nanoparticles can exceed the yield strength of silica at loads larger than ≈52 nN, and hence, it cannot be excluded that this dissipation mechanism contributes to the increase in friction, as well.

In our previous work,[21] we hypothesized that, upon approach of the tip/colloid to the surface at a velocity of 30 nm s^{-1}, the surface roughness (nanoparticles with surface densities $\rho = 170$ and 1100 μm^{-2}) led to a higher resistance against the squeeze-out of the IL ions compared to the smooth surface. This argument was used to justify the increase in the force required to resolve the IL layer closest to the surface. We can evaluate this hypothesis in light of our hydrodynamic force measurements, although very cautiously, since surface force measurements were carried out at much smaller shear rates than the hydrodynamic force measurements, which should cause a decrease of the slip length.[40] We have demonstrated that the resistance to flow through a high surface density of nanoparticles could bring the IL to rest. While this could happen on 1100 μm^{-2} substrates, our hydrodynamic force measurements do not support the increase in the layering force on the 170 μm^{-2} substrates. Therefore, we conclude that the observed increase in the layering force is not due to a hydrodynamic drag. Instead, we propose that the surface topology can template a different interfacial structure, in agreement with recent MD simulations.[20]

4. Conclusions

In the weakly adhesive contact between a silicon tip or a silica colloid and silicon wafers with modulated nanoscale roughness lubricated by [HMIM][TFSI], the lubrication mechanisms are influenced by the roughness of the substrate, and the roughness effects are dependent on the contact stress. If the stress is not high enough to eliminate the low-friction regime, friction achieves a minimum on the 275 μm^{-2} substrates. This is associated with the lower shear resistance of the IL along the surface as a result of the promoted slip, as illustrated by a slip length close to ≈100 nm. This mechanism has a limit since high nanoparticle surface densities enhance the viscous dissipation as fluid flows past surface irregularities and brings it to rest, regardless of how (weakly or strongly) the liquid molecules are attracted to the surface. At contact stresses above ≈200 MPa, the interfacial structure and the increased dissipation due to the surface roughness play a more important role. To the best of our knowledge, this is the first study that connects friction and hydrodynamic slip of an IL along surfaces with varying nanoscale roughness. Since the interaction strength between the liquid and the surface depends on the chemical composition of both of them, a prediction of the behavior of other ILs is not possible yet. However, the relevant mechanisms are expected to be universal. Furthermore, the results of this study along with our previous work corroborate that properties of the solid/IL interface such as interfacial structure, hydrodynamic slip, and friction coefficient cannot be directly extrapolated

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from ideally smooth to rough contacts. As a consequence, laboratory studies that represent better the real rough contact are needed both to understand the performance of ILs in applications and to design tribological systems.

5. Experimental Section

Ionic Liquid: [HMIM][TFSI] (purity 99%, Iolitec, Alabama, USA) was selected for this study. Figure S1 in the Supporting Information shows the molecular structure of cation and anion and their size, as determined by molecular mechanics with the Avogadro software (Version 1.0.3). [HMIM][TFSI] was equilibrated at 44% RH at room temperature for 10 days. The water uptake of [HMIM][TFSI] was determined gravimetrically to be ≈0.18 wt%. Force measurements were performed under ambient laboratory conditions (44–50% RH, 25 °C), i.e., a slight uptake of water during the experiments cannot be ruled out.

Preparation of Rough Substrates: Silicon wafers (p-type Boron ⟨111⟩ 500 μm, WRS, USA) were cut to 1 cm x 3 cm using a diamond pen. The cut silicon substrates were cleaned by ultrasonication in toluene, isopropanol, and ethanol (solvent from Sigma Aldrich, St. Louis, USA) in succession, three times in each solvent for 15 min. The rough substrates were prepared following the protocol in refs. [24–26]. Briefly, a solution of branched polyethyleneimine (PEI) with a molecular weight of 750 kDa (50 wt%, Sigma Aldrich, St. Louis, USA) was diluted to a concentration of 0.5 mg mL^{-1} in deionized water and stirred overnight at 50 °C to ensure dissolution of the polymer. The PEI solution was filtered with a 0.2 μm syringe filter twice before use. The wafers were UV-ozone treated for 30 min, then immersed for 30 min in the prepared PEI solution to allow adsorption of the polymer onto the wafer. Following adsorption of the polymer, the substrates were rinsed thoroughly with Milli-Q water. A suspension of silica nanoparticles (NPs) (5 wt% concentration, 12 nm nominal diameter, Microspheres-Nanospheres, NY, USA) was diluted in deionized water to a concentration of 0.002 wt% and ultrasonicated for 15 min directly before use. The PEI-coated wafer was then approached vertically to the NP suspension (≈45 mL), until one of the 1 cm edges was immersed into the solution (≈1 mm) and held for 5 min to eliminate the capillary effects in the rest of the wafer. Then, the entire wafer was completely submerged in the suspension and held for a set time between 15 and 25 min to achieve different number densities of surface-adsorbed NPs (given as number of NPs per μm^2). After the selected immersion times the small beaker containing the NP suspension and the wafer was placed in a larger beaker (≈300 mL) filled with Milli-Q water to dilute the NP suspension. The wafer was removed slowly from the solution.

Bare silicon wafers (as reference smooth surfaces) and the nanoparticle-coated surfaces were sintered up to 1080 °C using a heating ramp of 10 °C min^{-1}, followed by a cooling ramp of 2 °C min^{-1}. Sintering leads to an oxide layer on the silicon wafers;[41] they will be referred to as silica substrates in the following. The substrates were cleaned according to the previously described solvent cleaning method, and then UV-ozone treated for 30 min immediately before force measurements. After force measurements were performed, the substrates were immersed in acetone overnight and cleaned according to the above-described method before further use.

AFM Imaging: AFM images of rough and flat silica substrates were collected in AC mode at scan rates of 1 Hz in air with an MFP-3D (Asylum Research, California, USA) using gold-coated silicon cantilevers with a sharp tip of nominal radius of less than 10 nm, a resonant frequency of ≈300 kHz, and nominal spring constant of 20–75 N m^{-1} (Budget Sensors, Bulgaria). Tips were UV-ozone treated for at least 30 min prior to use.

The roughness of the silica substrates was characterized by the number of NPs per μm^2 in an area of 500 nm × 500 nm, which were counted manually on six images for each substrate. To determine the average distance between NPs, nine lines were considered on each image, and the average distance was determined from the scan-line size divided by the number of NPs. The surface density of NPs was estimated using this interparticle distance and assuming a face-centered cubic unit cell. Peak-to-valley and asperity-to-asperity distances were obtained by averaging at least ten profiles per image at three separate locations for each of the substrates. Several "smooth" sintered substrates (i.e., with an NP surface density of 0 per μm^2) were used in force measurements, while only two rough substrates were employed in force measurements—one that was immersed for 15 min in the suspension and another one that was immersed for 25 min—to investigate two statistically different characteristic roughness values.

Pull-Off and Friction Force Measurements: Force measurements were obtained using a NanoWizard 4a (JPK, Berlin, Germany) and either a silicon tip (CSC37/No Al, normal spring constant of 0.3–0.9 N m^{-1}, Mikromasch, Estonia) or silica colloids (Microspheres-Nanospheres, New York, USA) with nominal diameters of 10 μm. The silica colloids were glued with NOA 63 optical glue (Norland, New Jersey, USA) to tipless silicon rectangular cantilevers (CSC38/tipless/Al Bs, k = 0.3–0.9 N m^{-1}, Mikromasch, Mikromasch, Tallinn, Estonia). Immediately prior to the force measurements, the cantilevers were rinsed with ethanol, followed by 40 min of UV-ozone treatment. The normal spring constant of all AFM tips was determined by the thermal-noise method.[42] The lateral sensitivity (nN V^{-1}) of each tip was determined using the wedge calibration method[43] on a TGF11 calibration grid (MikroMasch, Estonia). The roughness and the radius of the colloidal probes were measured by reverse imaging the colloids on a TGT 1 test grating (NT-MDT, USA). Only colloids with RMS roughness (at the contact zone ≈0.5 × 0.5 um^2) smaller than 1.4 ± 1.1 nm were employed for the measurements.

A droplet of ≈100 μL of [HMIM][TFSI] was pipetted onto the substrate. After an equilibration time of at least 45 min, normal force measurements were carried out on 5 μm x 5 μm force maps with a total of 64 force measurements on each substrate to determine the average pull-off force and to confirm the presence of IL layers within the interfacial region, in agreement with the previous studies.[21] The approach speed was maintained constant at 30 nm s^{-1}. X–Y piezo velocity was 50 nm s^{-1} during force mapping to reduce noise.

Load-dependent friction force measurements with the sharp tip were obtained at the speed of 1 μm s^{-1} with loads ranging from 1 to 40 nN. Speed-dependent friction measurements were obtained using constant loads of 10 and 60 nN at sliding speeds ranging from 50 nm s^{-1} to 150 μm s^{-1}. Friction force measurements were also conducted with the colloidal probe as a function of the load (at constant sliding velocities of 5 μm s^{-1}) and of sliding velocity (at a load of 90 nN). Contact mode images in [HMIM][TFSI] were taken using the above described contact mode sharp tips to confirm that peaks seen in the lateral line scan correlated with the changes in height due to the presence of NPs. The number of peaks per 1 μm lateral line scan lines were manually counted and agreed well with the known average number of nanoparticles per 1 μm. After friction measurements, the tips were immersed in acetone overnight, then in isopropanol overnight, and finally in ethanol for at least 2 h. The tips were dried and treated with UV-ozone for 30 min before further use.

DMT Model: Because the Tabor parameter for both the silica colloids and the silicon tips was smaller than 0.1, the DMT model[29] was used to determine the contact area. The DMT model assumed perfectly smooth and elastic surfaces. The contact radius a could be obtained from

$$a^3 = \frac{3R}{4E^*}\left(F + 4\pi\gamma R\right) \tag{4}$$

where F is the applied force, R is the tip radius, γ *is* the surface energy, and E^* is the contact elastic modulus defined as

$$\frac{1}{E^*} = \left(\frac{1-\nu_{\text{sub}}^2}{E_{\text{sub}}}\right) / \left(\frac{1-\nu_{\text{tip}}^2}{E_{\text{tip}}}\right) \tag{5}$$

where E_{sub} is the elastic modulus of the sintered silica, E_{tip} is the elastic modulus of the probe, which was taken as ≈72.2 GPa for the substrate/silica colloid[44] and as 120 GPa for the sharp silicon tip, and ν is the Poisson's ratio of the substrate/colloid (ν_{sub} = 0.17) and the tip (ν_{tip} = 0.15), respectively.

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Hydrodynamic Force Measurements: The hydrodynamic force was measured by colloidal probe AFM on the selected substrates. Because the roughness of the colloid could affect the results, the same colloid was employed for the measurements on the three substrates. Two cantilevers were used: one with a spring constant of 0.204 N m^{-1} and a colloid with a radius of 4.0 μm and a second one with a spring constant of 0.391 N m^{-1} and a colloid with a radius of 2.2 μm. The cantilever was approached to the surface from a distance of ≈4 μm and a maximum load of 15 nN was applied. The approach velocity was ranged from 0.5 to 50 μm s^{-1}, and at least ten force curves were taken at each velocity at the same position. The measurements were carried out at room temperature.

To minimize thermal and mechanical drift, the samples and the cantilever were stabilized at least for 45 min in the IL with the cantilever a few microns away from the substrate. Out-of-contact dwell measurements were conducted in the IL to measure the change in cantilever deflection as a function of dwell time. The change in the deflection during the dwell tests was negligible in comparison to the change in deflection during hydrodynamic force measurements. The acquisition rates during hydrodynamic force measurements were carefully selected to measure enough data points even at the highest approach velocities. This enabled to obtain reliable fits of the model to the experimental results. A good fit of the model to the experimental results was achieved at separations larger than ≈20 nm. The deviations were attributed at smaller distances due to the interference with other interfacial phenomena. In the case of ILs, this was mainly due to the ordering of the IL ions at the solid surfaces and to the electrostatic interactions.

A nonlinear piezo extension (approach) might be a source of error in the approach velocity, a key input parameter in Equation (1). Hence, the actual tip velocity was determined with the actual piezo position and the cantilever deflection at each point of time, and it was found that it considerably differed from the set velocity. In contrast to other works,[45] using the actual velocity at each point of time did not lead to any instability of the solution. The hard wall of the approach curve was corrected with the individual compliance (instead of its average value) and the baseline (virtual deflection of the cantilever) was corrected at separations larger than 2 μm during post-processing.[46] Note that the reported surface separation accounted for the deflection of the cantilever. The hard wall was not always achieved at the highest velocities, which hindered the fits at the highest investigated velocities beyond 20 μm s^{-1}.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

adhesion, friction, ionic liquids, roughness, slip length

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[1] T. Welton, *Chem. Rev.* **1999**, *99*, 2071.
[2] C. Ye, W. Liu, Y. Chen, L. Yu, *Chem. Commun.* **2001**, 2244.
[3] a) Q.-G. Zhang, S.-S. Sun, S. Pitula, Q.-S. Liu, U. Welz-Biermann, J.-J. Zhang, *J. Chem. Eng. Data* **2011**, *56*, 4659; b) H. Weingärtner, *Angew. Chem., Int. Ed.* **2008**, *47*, 654; c) M. Armand, F. Endres, D. R. MacFarlane, H. Ohno, B. Scrosati, *Nat. Mater.* **2009**, *8*, 621; d) H. Liu, Y. Liu, J. Li, *Phys. Chem. Chem. Phys.* **2010**, *12*, 1685.
[4] a) M. Freemantle, *Chem. Eng. News* **1998**, *76*, 32; b) M.-D. Bermúdez, A.-E. Jiménez, J. Sanes, F.-J. Carrión, *Molecules* **2009**, *14*, 2888; c) M. Palacio, B. Bhushan, *Tribol. Lett.* **2010**, *40*, 247.
[5] R. Hayes, D. Wakeham, R. Atkin, in *Ionic Liquids Uncoiled* (Eds: N. V. Plechkova, K. R. Seddon), John Wiley & Sons, Inc., New York **2012**, p. 51.
[6] R. M. Espinosa-Marzal, M. Han, A. Arcifa, N. D. Spencer, A. Rossi, in *Encyclopedia of Interfacial Chemistry* (Ed: K. Wandelt), Elsevier, Oxford **2018**, p. 172.
[7] R. G. Horn, D. F. Evans, B. W. Ninham, *J. Phys. Chem.* **1988**, *92*, 3531.
[8] a) R. Hayes, G. G. Warr, R. Atkin, *Chem. Rev.* **2015**, *115*, 6357; b) M. V. Fedorov, A. A. Kornyshev, *Chem. Rev.* **2014**, *114*, 2978.
[9] a) R. Atkin, S. Z. El Abedin, R. Hayes, L. H. S. Gasparotto, N. Borisenko, F. Endres, *J. Phys. Chem. C* **2009**, *113*, 13266; b) B. Fitchett, J. Conboy, *J. Phys. Chem. B* **2004**, *108*, 20255; c) J. B. Rollins, B. D. Fitchett, J. C. Conboy, *J. Phys. Chem. B* **2007**, *111*, 4990; d) P. M. Richard, A. Claire, W. Tom, A. H. Patricia, *J. Phys.: Condens. Matter* **2014**, *26*, 284112; e) Z. Liu, T. Cui, T. Lu, M. Shapouri Ghazvini, F. Endres, *J. Phys. Chem. C* **2016**, *120*, 20224; f) K. Ueno, M. Kasuya, M. Watanabe, M. Mizukami, K. Kurihara, *Phys. Chem. Chem. Phys.* **2010**, *12*, 4066; g) A. M. Smith, K. R. Lovelock, N. N. Gosvami, P. Licence, A. Dolan, T. Welton, S. Perkin, *J. Phys. Chem. Lett.* **2013**, *4*, 378; h) A. M. Smith, K. R. J. Lovelock, S. Perkin, *Faraday Discuss.* **2013**, *167*, 279.
[10] a) R. Hayes, N. Borisenko, M. K. Tam, P. C. Howlett, F. Endres, R. Atkin, *J. Phys. Chem. C* **2011**, *115*, 6855; b) H. Li, F. Endres, R. Atkin, *Phys. Chem. Chem. Phys.* **2013**, *15*, 14624; c) H. Li, R. J. Wood, F. Endres, R. Atkin, *J. Phys.: Condens. Matter* **2014**, *26*, 284115.
[11] R. Hayes, G. G. Warr, R. Atkin, *Phys. Chem. Chem. Phys.* **2010**, *12*, 1709.
[12] a) P. K. Cooper, H. Li, M. W. Rutland, G. B. Webber, R. Atkin, *Phys. Chem. Chem. Phys.* **2016**, *18*, 23657; b) H. Li, M. W. Rutland, R. Atkin, *Phys. Chem. Chem. Phys.* **2013**, *15*, 14616; c) H. Li, R. J. Wood, M. W. Rutland, R. Atkin, *Chem. Commun.* **2014**, *50*, 4368; d) J. Sweeney, F. Hausen, R. Hayes, G. B. Webber, F. Endres, M. W. Rutland, R. Bennewitz, R. Atkin, *Phys. Rev. Lett.* **2012**, *109*, 155502; e) N. Hjalmarsson, D. Wallinder, S. Glavatskih, R. Atkin, T. Aastrup, M. W. Rutland, *Nanoscale* **2015**, *7*, 16039.
[13] O. Y. Fajardo, F. Bresme, A. A. Kornyshev, M. Urbakh, *Sci. Rep.* **2015**, *5*, 7698.
[14] a) H.-W. Cheng, P. Stock, B. Moeremans, T. Baimpos, X. Banquy, F. U. Renner, M. Valtiner, *Adv. Mater. Interfaces* **2015**, *2*, 1500159; b) H. W. Cheng, J. N. Dienemann, P. Stock, C. Merola, Y. J. Chen, M. Valtiner, *Sci. Rep.* **2016**, *6*, 30058; c) R. M. Espinosa-Marzal, A. Arcifa, A. Rossi, N. D. Spencer, *J. Phys. Chem. Lett.* **2014**, *5*, 179; d) L. A. Jurado, H. Kim, A. Rossi, A. Arcifa, J. K. Schuh, N. D. Spencer, C. Leal, R. H. Ewoldt, R. M. Espinosa-Marzal, *Phys. Chem. Chem. Phys.* **2016**, *18*, 22719; e) K. Sakai, K. Okada, A. Uka, T. Misono, T. Endo, S. Sasaki, M. Abe, H. Sakai, *Langmuir* **2015**, *31*, 6085.
[15] R. M. Espinosa-Marzal, A. Arcifa, A. Rossi, N. D. Spencer, *J. Phys. Chem. C* **2014**, *118*, 6491.
[16] A. M. Smith, K. R. J. Lovelock, N. N. Gosvami, T. Welton, S. Perkin, *Phys. Chem. Chem. Phys.* **2013**, *15*, 15317.
[17] M. Han, R. M. Espinosa-Marzal, *Lubricants* **2018**, *6*, 64.
[18] P. K. Cooper, C. J. Wear, H. Li, R. Atkin, *ACS Sustainable Chem. Eng.* **2017**, *5*, 11737.

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[19] A. C. F. Mendonça, A. A. H. Pádua, P. Malfreyt, *J. Chem. Theory Comput.* **2013**, *9*, 1600.
[20] A. David, O. Y. Fajardo, A. A. Kornyshev, M. Urbakh, F. Bresme, *Faraday Discuss.* **2017**, *199*, 279.
[21] A. Sheehan, L. A. Jurado, S. N. Ramakrishna, A. Arcifa, A. Rossi, N. D. Spencer, R. M. Espinosa-Marzal, *Nanoscale* **2016**, *8*, 4094.
[22] R. S. Anaredy, S. K. Shaw, *Langmuir* **2016**, *32*, 5147.
[23] M. Tariq, P. J. Carvalho, J. A. P. Coutinho, I. M. Marrucho, J. N. C. Lopes, L. P. N. Rebelo, *Fluid Phase Equilib.* **2011**, *301*, 22.
[24] S. N. Ramakrishna, R. M. Espinosa-Marzal, V. V. Naik, P. C. Nalam, N. D. Spencer, *Langmuir* **2013**, *29*, 15251.
[25] S. N. Ramakrishna, L. Y. Clasohm, A. Rao, N. D. Spencer, *Langmuir* **2011**, *27*, 9972.
[26] C. Huwiler, T. P. Kunzler, M. Textor, J. Voros, N. D. Spencer, *Langmuir* **2007**, *23*, 5929.
[27] M. Ruths, J. N. Israelachvili, in *Nanotribology and Nanomechanics II: Nanotribology, Biomimetics, and Industrial Applications* (Ed: B. Bhushan), Springer Berlin Heidelberg, Berlin, Heidelberg **2011**, p. 107.
[28] K. Xin, J. C. Lambropoulos, *Proc. SPIE* **2000**, *4102*, 112.
[29] K. L. Johnson, K. Kendall, A. Roberts, *Proc. R. Soc. London, Ser. A* **1971**, *324*, 301.
[30] D. Maugis, H. Pollock, *Acta Metall.* **1984**, *32*, 1323.
[31] M. Reitsma, V. Craig, S. Biggs, *Int. J. Adhes. Adhes.* **2000**, *20*, 445.
[32] O. Werzer, E. D. Cranston, G. G. Warr, R. Atkin, M. W. Rutland, *Phys. Chem. Chem. Phys.* **2012**, *14*, 5147.
[33] O. Werzer, R. Atkin, *Phys. Chem. Chem. Phys.* **2011**, *13*, 13479.
[34] T. D. Jacobs, R. W. Carpick, *Nat. Nanotechnol.* **2013**, *8*, 108.
[35] O. Reynolds, *Philos. Trans. R. Soc. London* **1886**, *177*, 157.
[36] Y. Zhu, S. Granick, *Phys. Rev. Lett.* **2002**, *88*, 106102.
[37] O. I. Vinogradova, *J. Colloid Interface Sci.* **1995**, *169*, 306.
[38] T. Lee, E. Charrault, C. Neto, *Adv. Colloid Interface Sci.* **2014**, *210*, 21.
[39] a) M. Han, R. M. Espinosa-Marzal, *J. Phys. Chem. C* **2018**, *122*, 21344; b) M. Han, R. M. Espinosa-Marzal, *ACS Appl. Mater. Interfaces* **2019**, *11*, 33465.
[40] Y. Zhu, S. Granick, *Phys. Rev. Lett.* **2002**, *88*, 106102.
[41] B. E. Deal, A. S. Grove, *J. Appl. Phys.* **1965**, *36*, 3770.
[42] J. L. Hutter, J. Bechhoefer, *Rev. Sci. Instrum.* **1993**, *64*, 1868.
[43] a) W. Liu, K. Bonin, M. Guthold, *Rev. Sci. Instrum.* **2007**, *78*, 063707; b) R. J. Cannara, M. Eglin, R. W. Carpick, *Rev. Sci. Instrum.* **2006**, *77*, 053701; c) F. Wang, X. Zhao, *Rev. Sci. Instrum.* **2007**, *78*, 043701; d) M. Varenberg, I. Etsion, G. Halperin, *Rev. Sci. Instrum.* **2003**, *74*, 3362; e) D. Ogletree, R. W. Carpick, M. Salmeron, *Rev. Sci. Instrum.* **1996**, *67*, 3298.
[44] W. Pabst, E. V. A. Gregorová, *Ceram.-Silik.* **2013**, *57*, 167.
[45] V. S. J. Craig, C. Neto, *Langmuir* **2001**, *17*, 6018.
[46] L. Zhu, P. Attard, C. Neto, *Langmuir* **2011**, *27*, 6712.

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